Lucas Gautheron

Computational Models of Human & Social Behavior

+33 6 79 23 86 47
lucas.gautheron@gmail.com
lucasgautheron.github.io
29/01/1995

Areas of Specialization **collective cognition, computational social science**

Education

May 2022 – Jul 2025 **Ph.D. in Sociology**, *Interdisciplinary Centre for Science and Technology Studies, Wuppertal*, Germany, *Summa cum laude*
"Social dilemmas in science: a computational approach to collective action problems and tradeoffs in collective cognition". Supervisors: Thomas Heinze & Radin Dardashti. Jury: Mark Lutter, Hugo Mercier, Camille Roth, Dunja Šešelja

2021-2022 **M.A., History and Philosophy of Science**, *Université Paris-Cité*, [Class rank: 1st]
Thesis : "Too Beautiful to be False, or Too Beautiful to be True? Searching for supersymmetry at the *Large Hadron Collider*". Supervisors: Olivier Darrigol & Elisa Omodei.

2014-2018 **Master 1, Theoretical and Experimental Physics**, *Ecole Normale Supérieure Paris-Saclay*

Research

Aug 2025 – Aug 2027 **Postdoctoral Fellow**, *Department of Philosophy, Evolution, Science and Society group, University of Missouri*

Mar 2023 – Dec 2024 **Research Engineer in Computational Linguistics (part-time)**, *Laboratoire de Sciences Cognitives et Psycholinguistique, Department of Cognitive Studies, École Normale Supérieure*, Paris
Investigating correlations between speech heard and produced by young infants.

Sep 2020 – Nov 2021 **Engineer in Computational Linguistics**, *Laboratoire de Sciences Cognitives et Psycholinguistique, Department of Cognitive Studies, École Normale Supérieure*, Paris
Language acquisition across cultures through naturalistic long-form audio recordings.

Oct 2016 – Jan 2017 **Research Internship (astrophysics)**, *Laboratoire Univers et Théories, CNRS*, Paris Meudon
Influence of the nuclei distribution on weak interactions' rates in core-collapse supernovae.

May 2016 – Jul 2016 **Research Internship (particle physics)**, *Laboratoire de Physique Nucléaire et des Hautes Énergies, CNRS*, Paris
Diphoton analysis and phenomenology for the ATLAS experiment.

May 2015 – Jul 2015 **Research Internship (particle physics)**, *Laboratoire d'Annecy-Le-Vieux des Particules, CNRS*, France
Particle physics for the ATLAS experiment.

Preprints

[A1] **L. Gautheron**, R. Marjieh, D. C. Conley, S. Frey, H. Rubin, M. D. Schneider, O. Tchernichovski, and N. Jacoby. *Popularity Feedback Constrains Innovation in Cultural Markets*. 2026.

[A2] **L. Gautheron**, N. Jacoby, and P. Harrison. *Active Inference with People: a general approach to real-time adaptive experiments*. 2026.

[A3] **L. Gautheron**. *Dilemmas and trade-offs in the diffusion of conventions*. 2025.

Peer-Reviewed Publications

[A4] **L. Gautheron**, E. Kidd, A. Malko, M. Lavechin, and A. Cristia. "Classification errors distort findings in automated speech processing: Examples and solutions from child-development research". In: *Behavior Research Methods* 58.6 (May 2026).

[A5] **L. Gautheron**. "Balancing specialization and adaptation in a transforming scientific landscape". In: *EPJ Data Science* 14.1 (Jan. 2025).

[A6] A. Cristia, **L. Gautheron**, Z. Zhang, B. Schuller, C. Scaff, C. Rowland, O. Räsänen, L. Peurey, M. Lavechin, W. Havard, C. Fausey, M. Cychosz, E. Bergelson, H. Anderson, N. Al Futaisi, and M. Soderstrom. "Establishing the reliability of metrics extracted from long-form recordings using LENA and the ACLEW pipeline". In: *Behavior Research Methods* (Sept. 2024).

[A7] **L. Gautheron** and E. Omodei. "How research programs come apart: The example of supersymmetry and the disunity of physics". In: *Quantitative Science Studies* 4.3 (Dec. 2023).

[A8] A. Cristia, **L. Gautheron**, and H. Colleran. "Vocal input and output among infants in a multilingual context: Evidence from long-form recordings in Vanuatu". In: *Developmental Science* 26.4 (Feb. 2023).

[A9] **L. Gautheron**, N. Rochat, and A. Cristia. "Managing, storing, and sharing long-form recordings and their annotations". In: *Language Resources and Evaluation* 57 (Feb. 2022).

[A10] M. Lavechin, M. de Seyssel, **L. Gautheron**, E. Dupoux, and A. Cristia. "Reverse Engineering Language Acquisition with Child-Centered Long-Form Recordings". In: *Annual Review of Linguistics* 8.1 (Jan. 2022).

Conference proceedings

[A11] L. Peurey, M. Lavechin, T. Kunze, M. Khentout, **L. Gautheron**, E. Dupoux, and A. Cristia. "Fifteen Years of Child-Centered Long-Form Recordings: Promises, Resources, and Remaining Challenges to Validity". In: *Interspeech 2025*. Aug. 2025.

Invited Talks

[T1] **L. Gautheron**. "A statistical physics approach to dilemmas and trade-offs challenging the diffusion of conventions". Institut Jean-Nicod, École Normale Supérieure, Paris. Mar. 2025.

[T2] **L. Gautheron**. ""When her family finds you are using the wrong metric...": dilemmas and trade-offs in the diffusion of scientific conventions". Department of Philosophy, London School of Economics, UK. Dec. 2024.

[T3] **L. Gautheron**. "Correlational analyses of the effects of caregivers' speech behaviour on children's speech production". Department of Linguistics, UCLA, Los Angeles, United States. June 2024.

[T4] **L. Gautheron**. "Algorithmic bias in correlational analyses of the effects of caregivers' speech behaviour on children's speech production". Daylong Audio Recordings of Children's Linguistic Environments (DARCLE), online. May 2024.

[T5] **L. Gautheron**. "Balancing Specialization and Adaptation in a Transforming Scientific Landscape: Modelling scientists' behavior with Natural Language Processing". NLP Seminar, LATTICE, Montrouge, France. Nov. 2023.

[T6] **L. Gautheron**. "Too beautiful to be false, or too beautiful to be true: supersymmetry and the future of high-energy physics". 2JM seminar, Sciences Po, Paris, France. May 2023.

[T7] **L. Gautheron** and A. Cristia. "A python package for long-form recordings and their annotations". Bergelson Lab, Duke University, NC, United States [online]. June 2021.

[T8] **L. Gautheron**. "Active inference with people: a general approach to real-time adaptive experiments". Department of Psychology, University of Missouri. Sept. 2025.

Selected Contributed Talks

[T9] **L. Gautheron**, R. Marjieh, D. C. Conley, S. Frey, H. Rubin, M. D. Schneider, O. Tchernichovski, and N. Jacoby. "Popularity Feedback Constrains Innovation in Cultural Markets". Annual Conference of the Cognitive Science Society (CogSci 2026). Upcoming (accepted). 9.3% acceptance rate. July 2026.

[T10] **L. Gautheron**, R. Marjieh, D. C. Conley, S. Frey, H. Rubin, M. D. Schneider, O. Tchernichovski, and N. Jacoby. "Popularity Feedback Constrains Innovation in Cultural Markets". 12$^{\text{th}}$ International Conference on Computational Social Science (IC2S2 2026). Upcoming (accepted). 34% acceptance rate. July 2026.

[T11] **L. Gautheron**. "Evolutionary mechanisms underlying herd behavior in science". Workshop "Scientific Progress via Model Transfer: The Case of Cultural Evolution"'. Apr. 2026.

[T12] E. Zanghi and **L. Gautheron**. "The Probability of Piety: Bayesian Perspectives on Cappadocian Church Decoration". 51st Byzantine Studies Conference (BSC) - Byzantine Studies Association of North America, Detroit. Oct. 2025.

[T13] L. Peurey, M. Lavechin, T. Kunze, M. Khentout, **L. Gautheron**, E. Dupoux, and A. Cristia. "Fifteen Years of Child-Centered Long-Form Recordings: Promises, Resources, and Remaining Challenges to Validity". In: *Interspeech 2025*. Aug. 2025.

[T14] **L. Gautheron**. "Dilemmas and trade-offs in the diffusion of conventions". 11th International Conference on Computational Social Science, Norrköping, Sweden. 25% acceptance rate. July 2025.

[T15] **L. Gautheron** and M. D. Schneider. "Plural pursuit across scales". Workshop on Large Language Models for the History, Philosophy, and Sociology of Science, TU Berlin. Apr. 2025.

[T16] **L. Gautheron**. "Social dilemmas in high-energy physics". Workshop "Methodological Transformations in Fundamental Physics", Wuppertal, Germany. Sept. 2024.

[T17] **L. Gautheron**. "A dialogue between philosophy of science and computational studies of science illuminates the crisis of fundamental physics". Workshop "Philosophy of Science meets Quantitative Science Studies"', Turin, Italy. May 2024.

[T18] **L. Gautheron**. "Probing Socio-Epistemic Dynamics in High-Energy Physics Using the Inspire HEP Database". Online conference "Big Data & History and Philosophy of Science". May 2023.

[T19] **L. Gautheron**. "The many faces of supersymmetry: Supersymmetry across subcultures of High-Energy Physics, 1971–2019". History of Science Society Annual Meeting, Chicago, United States. Nov. 2022.

[T20] **L. Gautheron**. "Who trusts supersymmetry? Probing quantitative methods for investigating research orientations in High-Energy Physics". 4th International Spring School of the Epistemology of the Large Hadron Collider: The History, Philosophy and Sociology of Large Scale Experiments, Wuppertal, Germany. Mar. 2022.

Poster presentations

[P1] **L. Gautheron**. "Balancing Specialization and Adaptation in a Transforming Scientific Landscape". 10th International Conference on Computational Social Science (IC2S2), Philadelphia, United States. July 2024.

Teaching, training, and learning material

Spring 2026 **Co-Instructor (10%) in Philosophy of Science**, *Graduate student seminar on the Philosophy of Big Science*, University of Missouri
Primary instructor: Mike D. Schneider

Spring 2025 **Co-Instructor (50%) in Computational Social Science**, *Seminar for masters' students and researchers*, École des hautes études en sciences sociales, Paris
Primary instructor: Camille Roth

Jul 2025 **Tutorial**, *Network analysis for historians, philosophers, and sociologists of science*, University of Wuppertal

Jan 2025 **Guest Lecturer in Philosophy**, *"Inverse problems for philosophers: Bridging the gap between agent-based models and behavioral data" [slides]*, Ruhr University Bochum
Primary instructor: Dunja Šešelja

Jun 2024 **Tutorial**, *"LongFoRMer: A software for managing and exploiting longform recordings and their annotations"*, UCLA, Department of Linguistics, Los Angeles, CA, United States

[B1] S. Pisani, L. Gautheron, and A. Cristia. *Long-form recordings: From A to Z*. 2021.

2015 **cosmology.education**, *Pedagogical website on the history of modern cosmology & science communication (7 500 users/year)*

2012 **Orbit**, *A C++ software for solving N-body problems, visualizing planetary motion, and predicting transits and eclipses.*, `https://orbit.sciencestechniques.fr/`

Research visits

Oct 2025 *Cornell University, Ithaca*

Feb 2023 – Aug 2023 **Visiting student**, *Medialab, Sciences Po Paris*

Jun 2022 *Max Planck Institute for Evolutionary Anthropology, Leipzig*

External collaborations

2025–2027 **"Designing Smart Environments to Augment Collective Learning & Creativity"**, *Member*
NSF project for studying the impact of self-governance on collective intelligence and creativity using large-scale experiments. PIs: Dalton Conley (Princeton) Seth Frey (UC Davis), Nori Jacoby (Cornell), and Ofer Tchernichovski (CUNY).

Graduate Workshops and Summer Schools attended

2025 **Santa Fe Institute Advanced Graduate Workshop in Computational Social Science Modeling and Complexity**, Santa Fe, NM, United States.

2024 **Santa Fe Institute Graduate Workshop in Computational Social Science Modeling and Complexity (20% acceptance rate)**, Santa Fe, NM, United States

2023 **Summer school "Collaboration and Interdisciplinarity in Science and Technology"**, Interdisciplinary Summer School Series in Higher Education Research and Science Studies, Wuppertal

2022 **Spring school "The History, Philosophy and Sociology of Large Scale Experiments"**, *4th International Spring School of the Epistemology of the Large Hadron Collider*, Wuppertal

Grants and awards

2026 **University of Missouri**, *CASCade: Building a Center for Applications in Social Complexity*, Co-Investigator, "Big-ideas" grant ($53 000)

2026 **University of Missouri**, MUPA travel award ($500)

2023 **G-Research**, Grants for PhD students and postdocs in quantitative fields (£1 000)

Service

Peer-review, ***Journals***: *Behavior Research Methods (×2); Philosophy of Science (×1).* ***Conferences***: *International Conference in Computational Social Science (×3)*

2024 **Workshop "Methodological Transformations in Fundamental Physics" (September 16th-18th)**, *Co-organizer*, Wuppertal

Journalism

Dec 2019 – Nov 2020 **President & Publication Director**, *Société de Production Le Média*, Montreuil
Head of a production company with 12+ full-time workers, a 1M€ annual budget, and 12 000 paid subscribers. Legal matters, revenue forecasts, audience data analysis, targeted marketing.

Sep 2018 – Sep 2020 **Journalist**, *Le Média*, Montreuil
Data journalism and book reviews, text & video (articles cited in research papers and books, and in an appeal to the French constitutional court).

Selected press articles

[M1] **L. Gautheron** and C. Gence. "[DATA] Les morts invisibles du coronavirus : la vérité derrière les chiffres officiels (The Hidden Toll of Coronavirus: Revealing the Truth Beyond Official Numbers)". In: *Le Média* (Apr. 4, 2020).

[M2] **L. Gautheron** and C. Gence. "[DATA] Lutte contre le COVID-19 : oui, la lenteur de l'État français a tué (Fatal Sluggishness: How France's COVID-19 Response Cost Lives)". In: *Le Média* (Apr. 23, 2020).

[M3] **L. Gautheron**. "Référendum ADP : les médias au service du pouvoir (Silencing Democracy: Media Blackout on the ADP Privatization Referendum)". In: *Le Média* (Aug. 1, 2019).

[M4] T. Kouamouo and **L. Gautheron**. "Élections européennes : un vote de classe avant tout (European Elections: It's Always About Class!)" In: *Le Média* (June 4, 2019).

[M5] **L. Gautheron**. "Des chiffres pour appréhender l'anti-Mélenchonisme de la presse". In: *Marianne* (Dec. 1, 2017).

Programming

2020 – On-going **LongFoRMer**, *A python package for managing and exploiting longform recordings and their annotations*, `https://github.com/LAAC-LSCP/ChildProject`

Jul 2013 **Back-end developer**, *Électricité réseau Distribution de France (ErDF)*, Annecy
Automated retrieval of large amounts of data from other company-wide applications.

2011 – 2014 **Volunteer**, *AssaultCube*
Development of a C++ 3D video game as part of an international team of volunteers.

Skills & interests

Programming: Python (expert), and C, C++, Fortran & Arduino (proficient); stan, sklearn, graph-tools, networkx, bertopic, pytorch; pandas, SQL; PHP, HTML, JS

Languages: French (native), English (fluent/C2); Spanish (beginner)

Interests: Energy transition; climate change and the polycrisis; microcontroller programming

References

Alejandrina Cristia (P.I., co-author)
Department of Cognitive Studies
École Normale Supérieure
Paris
alecristia@gmail.com
+33 (0)1 44 32 26 23
sites.google.com/site/acrsta

Radin Dardashti (Ph.D. advisor)
Interdisciplinary Center for Science and Technology Studies
Wuppertal, Germany
radin@gmx.de
+49 (0)202 439 2792
radindardashti.weebly.com

Olivier Darrigol (M.A. advisor)
M.A. advisor
CNRS
Université Paris-Cité
Paris
darrigol@paris7.jussieu.fr
sphere.univ-paris-diderot.fr

Elisa Omodei (M.A. advisor, co-author)
Department of Network and Data Science
Central European University
Vienna, Austria
omodeie@ceu.edu
people.ceu.edu/elisa_omodei

Thomas Heinze (Ph.D. advisor)
Interdisciplinary Center for Science and Technology Studies
Wuppertal, Germany
theinze@uni-wuppertal.de
+49 (0)202 439 2354
uni-wuppertal.de